Document Question Answering with Cited Sources

Pandas API Documentation Assistant

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1. Introduction

Large language models can produce fluent, confident answers even when they have no factual basis for them. This is especially costly in AI-assisted coding: developers routinely receive hallucinated function names and parameters that do not exist in their target library. This project addresses that problem by building a Retrieval-Augmented QA system that answers only from the official pandas documentation, cites the exact passage each answer came from, and explicitly refuses when the documentation does not contain the answer.

The system is evaluated on 120 hand-written questions split evenly into a development set (60) and a held-back set (60), with 40 answerable and 20 unanswerable questions in each. The unanswerable questions include fabricated parameters, wrong-library questions, and deprecated methods - the kinds of things a naive LLM would confidently hallucinate.

2. System Design

2.1 Pipeline Architecture

2.2 Components

Component	Model / Value
Embedding model	BAAI/bge-small-en-v1.5 (384-dim)
Vector index	FAISS IndexFlatIP (exact cosine similarity)
Generation model	Qwen/Qwen2.5-3B-Instruct

Component	Model / Value
Top-k	5
Chunk size	400 characters
Refusal threshold	0.82 (top passage similarity score)

2.3 Refusal Mechanism

The system uses a two-signal refusal:

1. **Score-based refusal:** If the top retrieved passage's similarity score is below 0.82, the system refuses without invoking the LLM.
2. **LLM-based refusal:** The LLM prompt instructs it to output NOT_FOUND when the passages do not contain the answer.

Both signals independently trigger refusal.

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3. Dataset

3.1 Document Corpus

- **Source:** pandas 3.0.5 API Reference Documentation
- **URL:** https://pandas.pydata.org/docs/reference/index.html
- **Licence:** BSD 3-Clause
- **Pages scraped:** 2,087
- **Passages after chunking (400 chars):** 11,855
- **Passages before chunking (raw):** 18,476

Each page is decomposed into up to four passage types: - **signature** (679 passages): function name + parameters - **description** (1,930 passages): what the function does - **parameters** (12,505 passages): each parameter explained - **examples** (3,362 passages): code snippets

3.2 Question Set

120 questions were hand-written before any system output was seen:

Category	Dev (60)	Heldback (60)	Total (120)
Answerable	40	40	80
Unanswerable - wrong library	5	5	10
Unanswerable - fake parameter	10	10	20

Category	Dev (60)	Heldback (60)	Total (120)
Unanswerable - deprecated method	5	5	10

Each answerable question records the answer AND the exact wording from the docs that supports it, enabling automated evaluation.

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4. Evaluation Setup

Six metrics computed by `src/evaluate.py`:

Metric	Definition	Direction
Retrieval recall@k	Fraction of answerable Qs with supporting wording in top-k	Higher = better
Answer correctness	Fraction of non-refused answers matching recorded answer	Higher = better
Unsupported answers	Fraction of non-refused answers not backed by cited passage	**Lower = better (main metric)**
Refuse when absent	Fraction of unanswerable Qs correctly refused	Higher = better
Refuse when present	Fraction of answerable Qs wrongly refused	Lower = better
Median response time	Median seconds per question	Lower = better

4.1 Heldback Discipline

The heldback set was locked with a `HELDBACK_LOCKED` marker file that made `evaluate.py` refuse to score against it. The lock was removed exactly once on Day 9 for the final evaluation. All tuning (chunk size, k, refusal threshold) was done on the dev set only.

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5. Configuration Studies (Days 5-8)

5.1 Day 5: Initial Baseline

Initial pipeline: chunk=800, k=5, no refusal threshold.

Metric	Value
Recall@k	77.5%
Answer correctness	61.5%
Unsupported answers	**39.3%**
Refuse when absent	90.0%
Refuse when present	35.0%
Response time	1.05s

5.2 Day 5: Citation Fix

Analysis showed the "unsupported" score was inflated because the citation only recorded one passage's text. The fix records all retrieved passages as the citation (representing what the LLM actually saw).

Result: unsupported dropped from 39.3% -> 21.2%.

5.3 Day 6: Configuration Sweep (3 x 3 = 9 experiments)

Systematically tested three chunk sizes and three k values:

chunk / k	k=3	k=5	k=10
400 chars	18.5%	**12.9%**	16.7%
800 chars	32.3%	21.2%	23.5%
1200 chars	34.6%	24.1%	15.8%

(Values shown are unsupported answer rate)

Winner: chunk=400, k=5 with 12.9% unsupported.

Insight: pandas parameter descriptions are short and distinct. Smaller chunks isolate each parameter, preventing interference.

5.4 Day 7: Alternative Retrievers (Negative Result)

Three alternative retrieval configurations tested:

Configuration	Recall	Unsupported
bge-small (Day 6 baseline)	80.0%	**12.9%**
bge-base	67.5%	16.1%

Configuration	Recall	Unsupported
bge-small + reranker (MS-MARCO)	67.5%	26.7%
bge-base + reranker	75.0%	21.2%

None beat the baseline. The MS-MARCO trained cross-encoder reranker was actively harmful because it was trained on web prose, not structured API documentation. The bigger bge-base model was also worse because bge-small handles keyword-heavy documentation text better despite being smaller.

This is a valuable negative result: "sophisticated" is not always "better", especially when training data mismatches the target domain.

5.5 Day 8: Refusal Threshold Sweep

Analysis of retrieval scores showed answerable questions typically had top scores > 0.7, while unanswerable questions had top scores < 0.6, but with substantial overlap.

A threshold sweep identified 0.82 as the best cut-off (constrained to keep refuse-when-absent >= 90%).

Metric	Before threshold	After threshold
Unsupported	12.9%	**7.1%**
Refuse when absent	90.0%	95.0%
Refuse when present	27.5%	32.5%

This was the biggest single improvement in the project.

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6. Failure Analysis (Day 9)

The Day 8 baseline was analyzed by categorizing every dev-set outcome:

Category	Count	%
A. Correct answer	14	23%
E. Correctly refused	19	32%
B. Wrongly refused	13	22%
C. Hallucinated	1	2%
D. Answer marked wrong	12	20%

Category	Count	%
F. Answer unsupported	1	2%

Total success rate: 55%; failure rate: 45%.

6.1 Root Cause of Category B (Wrongly Refused)

All 13 wrongly-refused questions had top retrieval scores between 0.79 and 0.87. Retrieval found the correct passages, but the LLM (Qwen 2.5 3B) still output NOT_FOUND. The model was being overly cautious despite having the answer in context.

6.2 Root Cause of Category D (Wrong Answer)

Manual inspection showed most Category D answers were factually correct but phrased differently from the recorded answer text. Example: - Recorded: "controls where NaN values are placed" - Model: "puts NaNs at the beginning if set to 'first'"

The evaluator's word-overlap threshold marks these as wrong even though a human would score them correct. This affects both v1 and Final pipelines equally.

6.3 Attempted Fix (Failed)

A "force-answer" rule was tested: if top retrieval score >= 0.80, override the LLM's NOT_FOUND response and provide an answer based on the top passage. Results on dev:

Metric	Day 8	With Fix
Unsupported	7.1%	**26.2% (worse)**
Refuse absent	95.0%	**75.0% (worse)**
Refuse present	32.5%	7.5%

The fix successfully reduced Category B but exploded Category C. Several unanswerable questions had high top scores (up to 0.93) because retrieval finds *topically similar* passages even when the docs contain no true answer. The fix was reverted.

Lesson: Retrieval score measures topic similarity, not answer presence. Single-signal refusal rules face inherent trade-offs.

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7. Final Results

7.1 Full Results Table

Metric	v1 Dev	v1 Heldback	Final Dev	Final Heldback
Retrieval recall@k	77.5%	72.5%	80.0%	72.5%
Answer correctness	50.0%	31.2%	55.6%	40.7%
Unsupported ▯	21.2%	15.6%	7.1%	14.8%
Refuse when absent	95.0%	100.0%	95.0%	100.0%
Refuse when present	20.0%	20.0%	32.5%	32.5%
Median time (sec)	1.11	1.33	1.28	0.53

7.2 Headline Findings

1. **The Final pipeline refuses ALL unanswerable questions on heldback** (100% refuse-when-absent). It never hallucinates on invented parameters or deprecated methods.

2. **Unsupported rate on heldback: 14.8%** (vs 15.6% for v1 baseline). The improvement is modest on heldback because the heldback split happened to be easier for v1 (15.6% vs 21.2% on dev), leaving less room for improvement.

3. **Dev-heldback gap: 7.7% on unsupported** (7.1% vs 14.8%). This indicates some overfitting to the dev set score distribution when choosing the 0.82 refusal threshold, but the final pipeline is still at least as good as v1 on every metric.

4. **Response time: 0.5-1.3 seconds** on Colab T4 GPU. Fast enough for interactive use.

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8. Discussion

8.1 What Worked

- **Aggressive chunking** (400 vs 800 chars) — big win on structured docs
- **Refusal threshold on retrieval score** — biggest single improvement
- **Combining LLM refusal + score refusal** — high refuse-when-absent
- **Storing supporting wording, not passage IDs** — evaluation survived chunk size changes

8.2 What Did Not Work

- **Larger embedding model (bge-base)** — worse on domain-specific text
- **MS-MARCO cross-encoder reranker** — actively harmful, wrong training domain
- **Force-answer override** — trades one error for another

8.3 Limitations

1. **Small evaluation set** (60 questions per split) introduces noise.
2. **Correctness metric** is strict — paraphrased correct answers score wrong.
3. **Single LLM tested** — larger models (7B, 13B) may have different refusal patterns.
4. **No retrieval score calibration** — scores mean different things at different retrieval qualities.

8.4 Future Work

1. **Multi-signal refusal:** combine LLM uncertainty, retrieval score, AND explicit answer-passage similarity check.
2. **Domain-adapted reranker:** fine-tune a cross-encoder on documentation-QA data.
3. **Semantic answer matching:** replace word-overlap correctness with an LLM-as-judge or embedding similarity metric.
4. **Larger evaluation set:** 500+ questions would reduce split noise.

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9. Reproducibility

All code, data, and results are in the git repository: https://github.com/mrerror313coder/pandas_qa

Key files: - `data/passages_400.jsonl` — passage corpus - `data/questions/dev.jsonl` — 60 dev questions - `data/questions/heldback.jsonl` — 60 heldback questions - `src/ingest.py` — scraping script - `src/embed_index.py` — index building - `src/run_pipeline.py` — end-to-end pipeline - `src/evaluate.py` — metrics computation - `results/BASELINE_predictions.jsonl` — final predictions - `results/HELDBACK_final_scores.json` — final scores

To reproduce:

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10. Conclusion

This project built a working RAG-based QA system for pandas API documentation. Over 10 days of systematic experimentation, the unsupported answer rate on the dev set was reduced from 39.3% to 7.1% through three targeted improvements: better citation handling, smaller chunks with k=5, and a retrieval-score refusal threshold.

The final pipeline achieves **100% refusal on unanswerable questions** in the held-back set - it never hallucinates when asked about fake parameters, wrong libraries, or deprecated methods. This is the core property the project set out to demonstrate: a QA system that knows what it does not know.

Configuration studies also produced valuable negative findings: alternative embedding models and cross-encoder rerankers made things worse due to domain mismatch. This suggests that for structured technical documentation, careful chunking and refusal calibration matter more than model size.
